

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 03

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Recap of Previous Lecture



Topic

Instruction

Topic

Multiple Instruction Support

Topics to be Covered



Topic

Variable Size Instructions

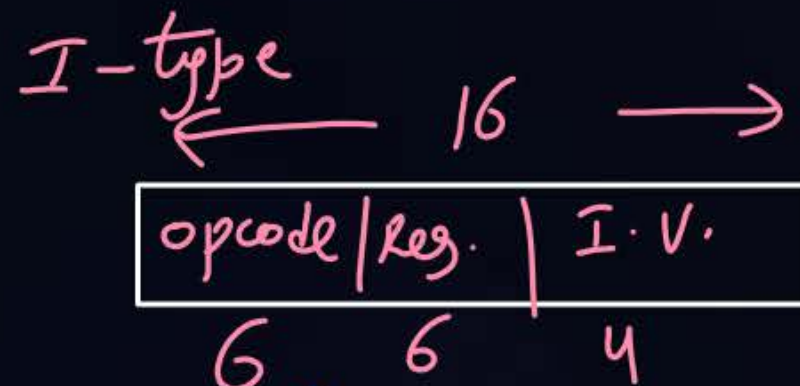
Topic

Constructing Instructions

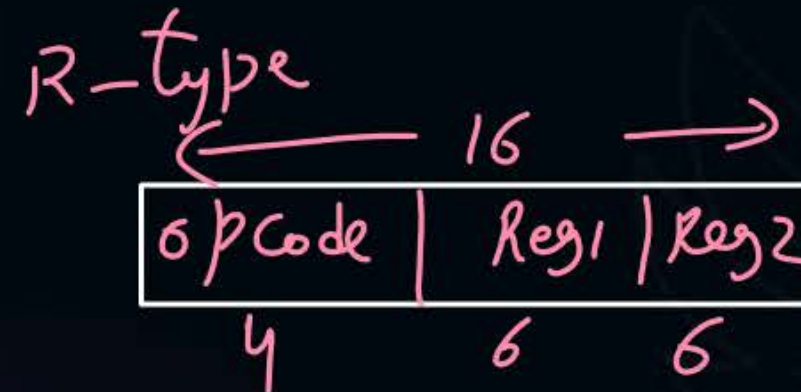
6 bits

Ans = 14

#Q. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____?



$(16-x) * 2^2 = 8$
 $16-x = 2$
 $x = 14$



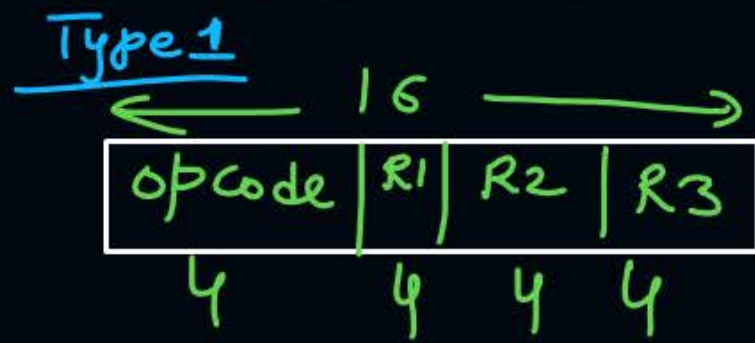
$\text{max} = 2^4 = 16$
 $\text{used} = x$
 $\text{unused} = 16 - x$

Ans = 32

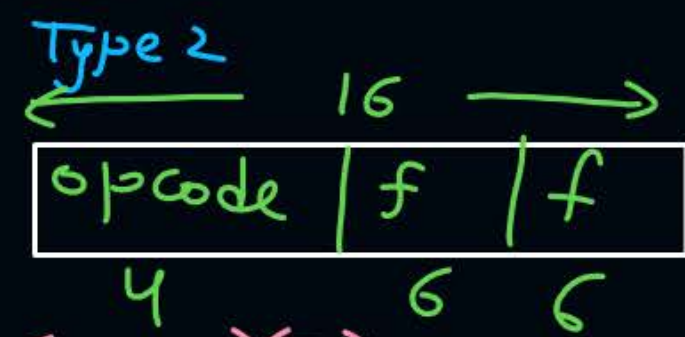
#Q. A processor has 16 integer registers ($R_0, R_1, \dots, R_{15}$) and 64 floating point registers ($F_0, F_1, \dots, F_{63}$). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type 4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating-point register operand (1F).

The maximum value of N is _____ ?

16 int reg. $\Rightarrow$ int reg = 4 bits | instⁿ size = 2 bytes = 16 bits
 64 float reg. $\Rightarrow$ float reg. = 6 bits



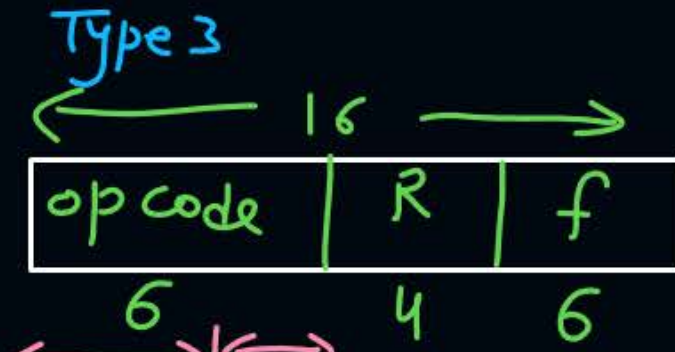
$$\begin{array}{r} \text{max} = 2^4 = 16 \\ \text{used} = 4 \\ \hline \text{unused} = 12 \end{array}$$



$\xleftarrow{4} \quad \xrightarrow{0}$

$$= 12 * 2^0 = 12$$

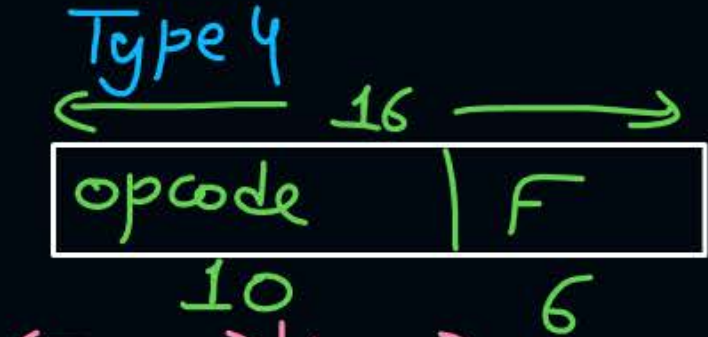
$$\begin{array}{r} \text{used} = 8 \\ \hline \text{unused} = 4 \end{array}$$



$\xleftarrow{4} \quad \xrightarrow{2}$

$$4 * 2^2 = 16$$

$$\begin{array}{r} \text{used} = 14 \\ \hline \text{unused} = 2 \end{array}$$



$\xleftarrow{6} \quad \xrightarrow{4}$

$$2 * 2^4 = 32$$

Instⁿ length

fixed length instruction



variable length opcode
(expanding opcode)

variable length instruction



fixed length opcode

#Q. Consider there are 3 types of instructions in system:

1. Register Operand instructions: One opcode and 2 registers
2. Memory Operand instructions: One opcode, 1 register and 1 memory address
3. Immediate Operand Instructions: One opcode, 1 register and 1 immediate operand

Number of registers = 64 $\Rightarrow$ Reg. operand = 6 bits

Number of bits in immediate operand = 10-bits

Memory size = 512Mbytes (byte addressable) $= 2^{29} \text{ B} \Rightarrow \text{add.} = 29 \text{ bits}$

Total Instructions:

1. Reg Operand type: 10
2. Memory Operand type : 12
3. immediate Operand type : 4

Total no. of inst^{ns} supported = $10 + 12 + 4 = 26$

$\Downarrow$
opcode bits = 5 bits

Maximum and Minimum instruction length are?

(40, 17) Ans.

Reg. operand



17 bits

mem. operand



40 bits

Imm. operand



21 bits

#Q. In a simplified computer the instructions are:

$\text{Reg}_i \leftarrow \text{Reg}_j$

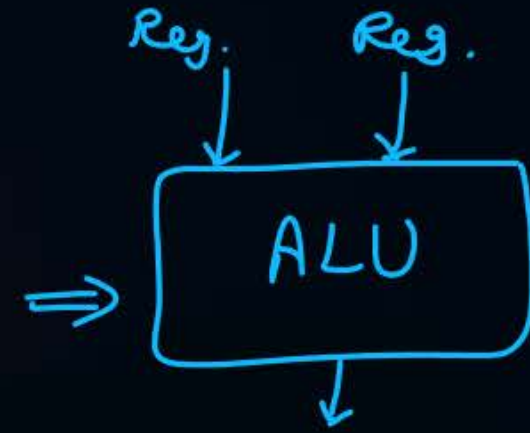
$\text{Reg}_i \leftarrow \text{Memory}$

$\text{Memory} \leftarrow \text{Reg}_i$

$\text{Reg}_i \leftarrow \text{Reg}_i \text{ op } \text{Reg}_j$

Reg.-based arch

2-add. inst^{ns}



The computer has only *four* registers and OP is either ADD or MUL.
Consider the following basic block:

$t1 = a + b$

$t2 = c + d$

$t3 = t1 * t2$

All operands are initially in memory. Construction instructions for the basic block with minimum number of instructions possible? *no. of inst^{ns} = 7*

$$R1 \leftarrow a$$

$$R2 \leftarrow b$$

$$R1 \leftarrow R1 + R2$$

$$R2 \leftarrow c$$

$$R3 \leftarrow d$$

$$R2 \leftarrow R2 + R3$$

$$R1 \leftarrow R1 * R2$$

#Q. In a simplified computer the instructions are:

OP R_i, R_j	- Performs $R_i \text{ Op } R_j$ and stores the result in R_j
OP m, R_i	- Performs $\text{val Op } R_i$ and stores the result in R_i val denotes the content of memory location m
MOV m, R_i	- Moves the content of memory location m to register R_i
MOV R_i, m	- Moves the content of register R_i to memory location m

} → Reg-mem.
based archit.

The computer has only two registers and OP is either ADD or SUB. Consider the following basic block:

$$t1 = a + b$$

$$t2 = c + d$$

$$t3 = e - t2$$

$$t4 = t1 - t3$$

#Q. Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

- | | |
|----------------------------|-------------------------|
| ① <code>MOV b, R1</code> | $R1 \leftarrow b$ |
| <code>ADD a, R1</code> | $R1 \leftarrow a + R1$ |
| ② <code>MOV d, R2</code> | $R2 \leftarrow d$ |
| <code>ADD c, R2</code> | $R2 \leftarrow c + R2$ |
| <code>SUB e, R2</code> | $R2 \leftarrow e - R2$ |
| <code>SUB R1, R2</code> | $R2 \leftarrow R1 - R2$ |
| ③ <code>MOV R2, mem</code> | $mem \leftarrow R2$ |

Ans = 5

#Q. In a simplified computer the instructions are:

OP R_i, R_j	- Performs $R_i \text{ Op } R_j$ and stores the result in R_i
OP R_i, m	- Performs $R_i \text{ Op val}$ and stores the result in R_i val denotes the content of memory location m
MOV m, R_i	- Moves the content of memory location m to register R_i
MOV R_i, m	- Moves the content of register R_i to memory location m

The computer has only two registers and OP is either ADD or SUB. Consider the following basic block:

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#Q. Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

① $R1 \leftarrow a$
 $R1 \leftarrow R1 + b$

② $R2 \leftarrow c$
 $R2 \leftarrow R2 + d$

③ $m \leftarrow R2$

④ $R2 \leftarrow e$
 $R2 \leftarrow R2 - m$
 $R1 \leftarrow R1 - R2$

⑤ $mem \leftarrow R1$

Register spill



Topic : Register Spill



when limited no. of reg.^s are there in CPU and if for temporary basis reg. value to be moved out to mem. .



#Q. Consider a register-memory architecture system (2-address instructions). For this system the following intermediate code is going to be converted in machine code. Minimum how many registers are required in system so that the code can run without register spill?

Consider first operand is always the register operand and it's the destination for operation too.

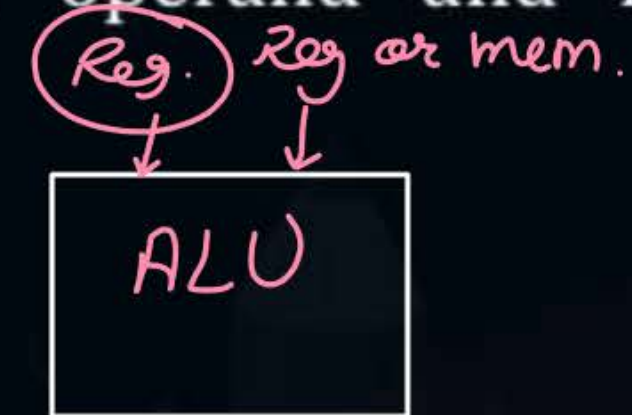
$t1 = X + Y$

$t2 = t1 - Z$

$t3 = t1 + t2$

$t4 = M + t3$

Assume X, Y, Z and M are memory operands



#Q. Consider a register-based architecture system (2-address instructions). For this system the following intermediate code is going to be converted in machine code. Minimum how many registers are required in system so that the code can run without register spill?

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$$t1 = X + Y$$

$$t2 = t1 - Z$$

$$t3 = t1 + t2$$

$$t4 = M + t3$$

Assume X, Y, Z and M are memory operands



2 mins Summary



Topic

Variable Size Instructions

Topic

Constructing Instructions





Happy Learning

THANK - YOU

